

NURSING CASE STUDY & CARE PLAN ASSIGNMENT

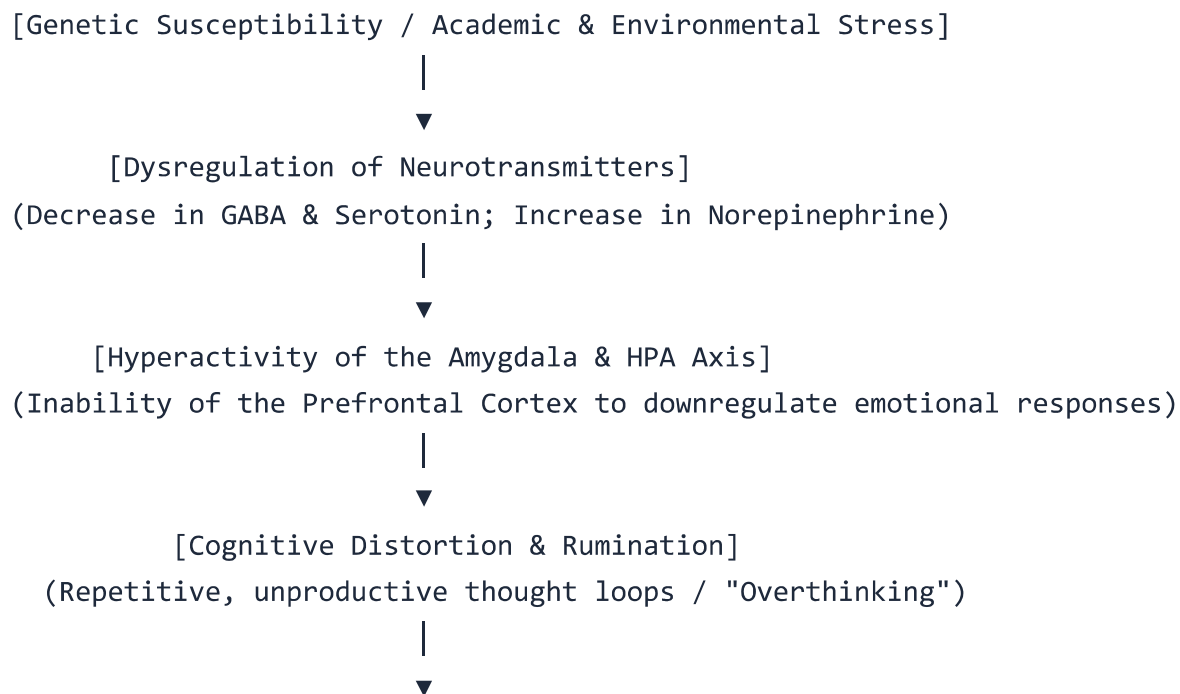
- **Subject:** Psychiatric-Mental Health Nursing (GNM Nursing Syllabus)
 - **Clinical Facility:** Adesh Medical College & Hospital, Mohri, Shahabad
 - **Patient Diagnosis:** Generalized Anxiety Disorder (GAD) with Obsessive Rumination
 - **Academic Level:** GNM (General Nursing and Midwifery)
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1. Introduction: Pathophysiology of the Disease

Generalized Anxiety Disorder (GAD) with Obsessive Rumination is a psychiatric condition characterized by excessive, uncontrollable, and irrational worry about everyday things, coupled with repetitive, prolonged, and passive thinking about one's distress, its causes, and its consequences (rumination), rather than active problem-solving.

In a clinical context, a GNM nursing student experiencing high academic pressure can develop maladaptive cognitive patterns leading to severe distress, sleep deprivation, and functional impairment.

Pathophysiological Mechanism (Flowchart Concept)



[Physiological & Psychological Symptoms]

(Palpitations, insomnia, muscle tension, severe anxiety)

- **Neurochemical Factors:** Decreased levels of Gamma-Aminobutyric Acid (GABA) and Serotonin reduce the brain's ability to inhibit stress responses. Elevated norepinephrine stimulates the sympathetic nervous system, inducing physiological anxiety symptoms.
- **Neuroanatomical Factors:** Hyperactivity in the amygdala (emotional hub) combined with hypoactivity in the prefrontal cortex (executive hub) leads to an inability to suppress negative ruminative cycles.
- **Cognitive-Behavioral Theory:** Rumination is used as a maladaptive coping mechanism to gain control over stress but results in a self-reinforcing loop of anxiety and cognitive exhaustion.

2. Patient Profile

Profile Parameter	Patient Details
Patient Name	Mr. Amit Kumar (*)
Age / Gender	21 Years / Male
Marital Status	Single
Occupation	Student
Education Status	GNM (General Nursing and Midwifery) Student
Religion	Hindu
Address	VPO Rasina, Kaithal, Haryana
Hospital Name	Adesh Medical College & Hospital, Mohri, Shahabad
Ward / Bed Number	Psychiatric Ward / Bed No. 2
Admission Date	2026-06-13
Medical Diagnosis	Generalized Anxiety Disorder (GAD) with Ruminative Features

Profile Parameter	Patient Details
Surgical Diagnosis	N/A (No surgical intervention indicated)

(*) AI Generated Academic Case Data for clinical simulation purposes.

3. Chief Complaints

The patient was admitted to the Psychiatric Ward with the following complaints, as reported by the patient and his family:

- **Persistent overthinking and repetitive thoughts** (rumination) regarding academic performance, exam failure, and future security for the last 6 months.
- **Difficulty falling and maintaining sleep** (onset insomnia) for the last 4 months.
- **Generalized restlessness, palpitations, and muscle tension** (specifically in the neck and shoulders) during high-stress periods.
- **Decreased concentration and memory retention**, severely impacting his nursing studies.
- **Frequent bouts of fatigue and social withdrawal** from classmates and family.

(*) AI Generated Academic Case Data.

4. Present History

Mr. Amit Kumar, a 21-year-old GNM nursing student, was apparently well 6 months ago. With the approaching university examinations and intensive clinical postings, he began experiencing progressive anxiety. He reported that he could not "switch off" his brain, constantly overthinking his clinical performance and mock test grades.

Over the past 3 months, these thoughts evolved into severe rumination, where he would spend 4 to 5 hours daily reviewing his perceived failures. His sleep deteriorated to less than 4 hours of fragmented sleep per night. He experienced somatic anxiety symptoms (palpitations, hand tremors, and epigastric discomfort).

Recognizing a steep decline in his academic performance and social interaction, his family consulted the outpatient psychiatric department at Adesh Medical College, leading to his voluntary admission on 2026-06-13 for diagnostic evaluation, therapy, and pharmacological stabilization.

(*) AI Generated Academic Case Data.

5. Past Medical & Surgical History

- **Past Medical History:** No prior history of diagnosed psychiatric illnesses, panic attacks, depressive episodes, or chronic systemic medical disorders (such as diabetes, hypertension, or thyroid dysfunction).
- **Past Surgical History:** No history of major surgical interventions or hospitalizations.

(*) AI Generated Academic Case Data.

6. Personal History

- **Diet:** Vegetarian. Currently reports decreased appetite (hyporexia) due to chronic anxiety.
- **Sleep Pattern:** Restless, fragmented sleep. Sleep latency (time taken to fall asleep) has increased to 1.5–2 hours. Total sleep duration is less than 5 hours of non-restorative sleep, despite spending 7+ hours in bed trying to sleep.
- **Elimination Pattern:** Normal bowel and bladder habits; occasional reports of constipation linked to high-stress periods.
- **Addictions:** None. Denies use of tobacco, alcohol, cannabis, or excessive caffeine.
- **Exercise/Activity:** Sedentary lifestyle. No active physical exercises due to excessive mental fatigue.
- **Hygiene:** Maintains good personal hygiene (bathes daily, well-groomed, wears clean clothes).

(*) AI Generated Academic Case Data.

7. Family History

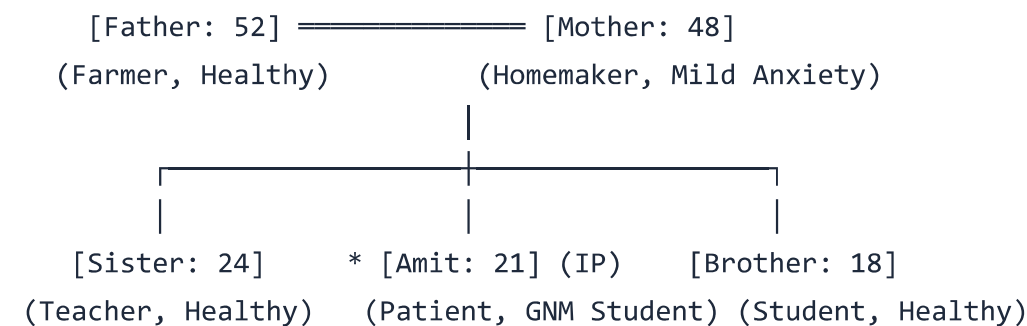
Mr. Amit Kumar belongs to a middle-class nuclear family consisting of 5 members. He is the second of three siblings.

- His father (52) is a farmer, and his mother (48) is a homemaker.

- There is a history of mild, untreated generalized anxiety in his mother, but no history of major psychiatric disorders, suicide attempts, or substance abuse in the immediate or extended family.
- The family is highly supportive but lacks a deep understanding of mental health conditions, initially dismissing his symptoms as "exam stress."

(*) AI Generated Academic Case Data.

8. Family Tree (Genogram)



Key:

- IP = Index Patient (Mr. Amit Kumar)
- [] = Male
- () = Female
- == = Marriage/Union
- — = Descent/Siblings

9. Physical Assessment

General & Anthropometric Assessment

- **General Appearance:** Alert, cooperative, but visibly anxious. Frequent sighing, fidgeting of hands, and poor eye contact during clinical assessment.
- **Height:** 172 cm
- **Weight:** 61 kg

- **Body Mass Index (BMI):** 20.6 kg/m² (Normal Range: 18.5–24.9 kg/m²)

Vital Signs (Recorded on Admission: 2026-06-13)

Vital Sign	Patient Value	Normal Value	Interpretation
Body Temperature	98.4 °F (Oral)	98.6 °F (Oral)	Normal (Apyrexial)
Pulse Rate	92 beats/minute	60–100 beats/minute	Within Normal Limits (Mildly tachycardic)
Respiratory Rate	20 breaths/minute	12–20 breaths/minute	Within Normal Limits (Upper border)
Blood Pressure	126/82 mmHg	120/80 mmHg	Pre-hypertension (likely stress-induced)

(*) AI Generated Academic Case Data.

10. Systemic Examination & Mental Status Examination (MSE)

Systemic Review

- **Cardiovascular:** S1, S2 heard. Mild tachycardia present during anxiety spikes. No murmurs.
- **Respiratory:** Clear bilateral lung fields. No wheezing, crackles, or rhonchi.
- **Gastrointestinal:** Soft, non-distended abdomen. Normal bowel sounds. Subjective complaint of "acid reflux" during stressful episodes.
- **Nervous & Musculoskeletal:** Cranial nerves intact. Deep tendon reflexes (DTR) normal (2+). Increased muscle tension noted in the trapezius and neck muscles.

Mental Status Examination (MSE) Summary

MSE Domain	Observed Clinical Finding
Appearance & Behavior	Neatly dressed, cooperative, posture is tense. Trembling fingers and frequent tapping of feet.

MSE Domain	Observed Clinical Finding
Speech	Rapid rate, normal volume, productive but anxious tone.
Mood and Affect	Mood described as "stressed and tired"; affect is anxious, congruent with mood, with restricted range.
Thought Process	Circumstantial, coherent, but exhibits persistent ruminations and overthinking regarding exams.
Thought Content	No delusions, no suicidal ideations. Preoccupied with academic failure.
Perception	No auditory or visual hallucinations. No illusions.
Cognitive Functions	Orientation: Oriented to Time, Place, and Person. Attention & Concentration: Impaired (struggles with serial 7 subtraction test). Memory: Intact immediate, recent, and remote memory.
Insight & Judgment	Insight: Grade IV (Social Insight - recognizes he is ill but blames environment/exams). Judgment: Intact personal and social judgment.

(*) AI Generated Academic Case Data.

11. Investigation Table

Investigation	Patient Value	Normal Reference Range	Clinical Interpretation
Hemoglobin (Hb)	14.2 g/dL	13.5–17.5 g/dL	Normal
Total Leukocyte Count	7,200 /mm ³	4,000–11,000 /mm ³	Normal (No infectious process)
Serum TSH	2.1 mIU/L	0.4–4.0 mIU/L	Normal (Rules out hyperthyroidism)
Serum Creatinine	0.9 mg/dL	0.6–1.2 mg/dL	Normal Kidney Function

Investigation	Patient Value	Normal Reference Range	Clinical Interpretation
Serum Glutamic Oxaloacetic Transaminase (SGOT)	24 U/L	5–40 U/L	Normal Liver Function
Electrocardiogram (ECG)	Sinus Tachycardia	Normal Sinus Rhythm	Heart rate 94 bpm, no ST-T segment abnormalities; consistent with acute anxiety.

(*) AI Generated Academic Case Data.

12. Drug Study

The following pharmacological regimen was prescribed by the psychiatrist at Adesh Medical College & Hospital on admission:

Drug Name (Generic/Trade)	Dosage & Route	Classification & Action	Side Effects	Nursing Responsibilities
Escitalopram (Nexito)	10 mg, PO, Once daily (OD) at night	SSRI (Selective Serotonin Reuptake Inhibitor): Blocks serotonin reuptake, increasing serotonin levels in synaptic clefts to reduce anxiety.	Nausea, dry mouth, headache, initial insomnia, sexual dysfunction.	• Administer with food to reduce GI upset. • Monitor for suicidal ideation, especially in the first 2 weeks of therapy. • Advise patient not to stop the drug abruptly (risk of withdrawal).

Drug Name (Generic/Trade)	Dosage & Route	Classification & Action	Side Effects	Nursing Responsibilities
Clonazepam (Lonazep)	0.25 mg, PO, Twice daily (BD) (PRN/Short- term)	Benzodiazepine / Anxiolytic: Enhances GABA activity, producing a calming effect on the CNS.	Drowsiness, dizziness, ataxia, risk of dependence/addiction.	• Monitor level of sedation and coordination. • Warn the patient against driving or operating machinery. • Strict monitoring to prevent psychological/physical dependence.
Multivitamin Supplement (Becosules)	1 Capsule, PO, OD after lunch	Nutritional Supplement: Replenishes Vitamin B- Complex and Vitamin C to combat physical stress and fatigue.	Mild gastric irritation, yellow coloration of urine.	• Administer after meals. • Reassure patient that yellow discoloration of urine is harmless and normal.

(*) AI Generated Academic Case Data.

13. Nursing Diagnoses

Based on NANDA-I guidelines, the following diagnoses have been prioritized for Mr. Amit Kumar:

Anxiety related to excessive academic stressors, fear of failure, and neurochemical imbalance, as evidenced by persistent ruminations, palpitations, trembling fingers, and a elevated pulse rate of 92 bpm.

Disturbed Thought Processes (Rumination) related to cognitive distortions and ineffective coping mechanisms, as evidenced by repetitive self-deprecating thoughts and decreased concentration during MSE.

Insomnia related to cognitive hyperactivity, pre-sleep worry, and physiological tension, as evidenced by a prolonged sleep onset latency (1.5–2 hours) and feeling unrefreshed upon waking.

14. Nursing Care Plan (NCP) Table

Diagnosis 1: Anxiety related to academic stressors and fear of failure as evidenced by persistent ruminations and somatic symptoms (palpitations, trembling).

NURSING CARE PLAN (DIAGNOSIS 1)

ASSESSMENT

- Subjective: Patient states, "I cannot stop my mind from overthinking about exams, and my chest feels tight when I think of failing."
- Objective: Pulse 92 bpm, fine hand tremors present, constant pacing in ward

NURSING DIAGNOSIS

- Anxiety related to academic stressors and fear of failure as evidenced by persistent ruminations and somatic symptoms (palpitations, trembling).

GOALS & OUTCOMES

- Short-term: Patient will report a subjective reduction in anxiety from severe to moderate within 48 hours.
- Long-term: Patient will demonstrate 2 adaptive relaxation techniques and maintain a normal heart rate (60-100 bpm) by discharge.

INTERVENTIONS

1. Establish a therapeutic, calm, and non-judgmental environment.
2. Instruct and guide the patient in deep-breathing exercises (4-7-8 technique) and progressive muscle relaxation (PMR).
3. Encourage the patient to express his fears and correct cognitive distortions regarding academic failure.
4. Administer prescribed Escitalopram and Clonazepam as ordered, monitoring for therapeutic and adverse effects.

RATIONALE

1. A calm environment reduces external stimuli that can exacerbate anxiety.
2. Deep breathing stimulates the vagus nerve, activating the parasympathetic nervous system to lower heart rate and muscle tension.

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| 3. Verbalization helps externalize internal distress, allowing the patient to analyze his thoughts logically.
4. Pharmacotherapy addresses neurochemical imbalances (GABA and serotonin). |
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EVALUATION

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| <ul style="list-style-type: none"> • Outcome Met: The patient's heart rate decreased to 78 bpm. He reported feeling calmer and was observed practicing deep-breathing exercises. |
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Diagnosis 2: Insomnia related to cognitive hyperactivity as evidenced by prolonged sleep latency and fragmented sleep.

NURSING CARE PLAN (DIAGNOSIS 2)

ASSESSMENT

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| <ul style="list-style-type: none"> • Subjective: Patient reports, "I lie awake for hours thinking about what I need to study. I wake up exhausted." • Objective: Dark circles under eyes, frequent yawning, dull facial expression. |
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NURSING DIAGNOSIS

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| <ul style="list-style-type: none"> • Insomnia related to cognitive hyperactivity as evidenced by prolonged sleep latency and fragmented sleep. |
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GOALS & OUTCOMES

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| <ul style="list-style-type: none"> • Short-term: Patient will verbalize understanding of sleep hygiene rules within 24 hours. • Long-term: Patient will report sleeping 6–7 hours of continuous, restorative sleep within 1 week. |
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INTERVENTIONS

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| 1. Teach the patient strict Sleep Hygiene Rules (avoid screen time 1 hour before bed, maintain a regular sleep schedule, avoid caffeine after 4 PM)
2. Implement a "Worry Time" technique: Allocate 15–20 minutes in the afternoon to write down worries, leaving the evening free of rumination.
3. Ensure a quiet, dark, and comfortable temperature in the ward room at night. Encourage warm milk or a warm bath before bed.
4. Administer Escitalopram in the morning or at night as tolerated, and |
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PRN Clonazepam before bedtime if severe insomnia persists.
RATIONALE
1. Cognitive interventions and environmental adjustments address sleep-onset barriers without relying solely on sedatives.
2. Worry time sets clear boundaries for anxious thoughts, preventing them from intruding into sleep hours.
3. Melatonin production is optimized in a dark, quiet, and comfortable environment, facilitating physiological sleep.
4. Appropriate timing of medications minimizes drug-induced insomnia or excessive daytime drowsiness.
EVALUATION
• Outcome Partially Met: Patient reported sleeping 5.5 hours continuously on the third night and was able to explain at least 4 sleep hygiene rules.

(*) AI Generated Academic Case Data.

15. Health Education

As a nursing student himself, Mr. Amit Kumar was educated using clinical, peer-to-peer academic language, which helped improve his compliance and insight.

1. Medication Compliance

- **Adherence:** Emphasized that **Escitalopram** takes 2 to 4 weeks to demonstrate full therapeutic efficacy. He must not discontinue the medication even if he feels better.
- **Avoid Self-Medication:** Instructed him on the high risk of dependency associated with **Clonazepam** and advised him to use it only for short-term relief as prescribed.

2. Cognitive & Lifestyle Modifications

- **Thought Stopping Technique:** Taught him to mentally say "Stop!" or gently snap a rubber band on his wrist when he notices himself entering a rumination loop.

- **Structured Studying:** Advised dividing study material into smaller, manageable chunks (Pomodoro Technique: 25 minutes of study, 5 minutes of rest) rather than continuous, high-pressure study sessions.
- **Avoid Stimulants:** Restrict caffeine consumption (tea, coffee) to morning hours to avoid worsening anxiety and insomnia.

3. Stress Management & Relaxation

- Daily practice of **Jacobson's Progressive Muscle Relaxation (JPMR)** for 15 minutes.
- Engaging in light cardiovascular exercise (e.g., 20-minute brisk walk in the evening) to help burn excess cortisol and adrenaline.

4. Follow-Up Plan

- Schedule follow-up consultations in the psychiatry outpatient department (OPD) every 2 weeks initially.
- Advised to seek immediate medical help if he experiences worsening anxiety, severe panic attacks, or thoughts of self-harm.

(*) AI Generated Academic Case Data.

16. Conclusion

Mr. Amit Kumar, a 21-year-old GNM nursing student, was admitted to the Psychiatric Ward of Adesh Medical College & Hospital with Generalized Anxiety Disorder characterized by severe academic overthinking and rumination. Through a combined approach of pharmacological management (SSRI and short-term benzodiazepine), cognitive interventions (sleep hygiene, thought stopping, worry-time therapy), and empathetic nursing care, the patient demonstrated significant improvement.

By the end of his psychiatric admission, his physical symptoms of anxiety subsided, his sleep quality improved, and he acquired adaptive coping mechanisms to manage academic stress. He was discharged in a stable state with advice for regular follow-up and continued psychotherapy.

(*) AI Generated Academic Case Data.